Secant function

1

a

x	$-\frac{\pi}{4}$	0	$-\frac{\pi}{6}$	$\frac{2\pi}{3}$	π	$\frac{5\pi}{4}$
$\sec x$	1.414	1	1.155	-2	-1	-1.414

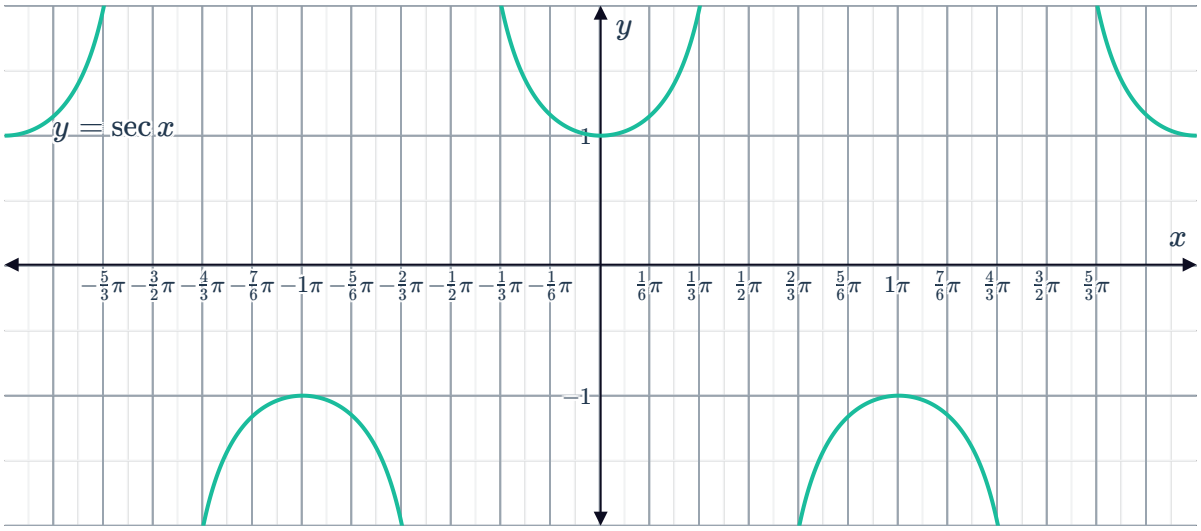
b $x = -\frac{3\pi}{2}, -\frac{\pi}{2}, \frac{\pi}{2}, \frac{3\pi}{2}$

c $x = -2\pi, 0, 2\pi$

d $x = -\pi, \pi$

e 2π

f



g $\left(\pi, \frac{3\pi}{2}\right), \left(\frac{3\pi}{2}, 2\pi\right)$

2

a

x	57.30°	85.94°	89.38°	89.95°	90.53°	91.67°	114.59°
$\sec x$	1.85	14.12	92.41	1145.92	-108.11	-34.31	-2.40

b The values of $\sec x$ get large very quickly. Given that $\cos 90^\circ = 0$, $\sec x$ approaches an asymptote at $x = 90^\circ$.

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3

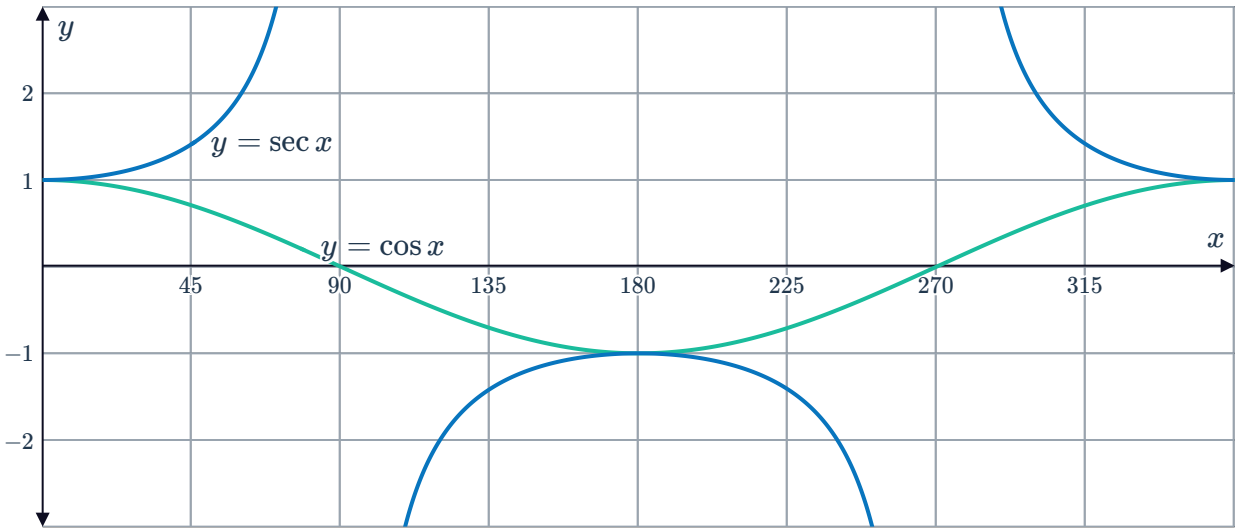
a

x	0°	45°	90°	135°	180°	225°	270°	315°	360°
$\cos x$	1	$\frac{1}{\sqrt{2}}$	0	$-\frac{1}{\sqrt{2}}$	-1	$-\frac{1}{\sqrt{2}}$	0	$\frac{1}{\sqrt{2}}$	1
$\sec x$	1	$\sqrt{2}$	–	$-\sqrt{2}$	-1	$-\sqrt{2}$	–	$\sqrt{2}$	1

b 1

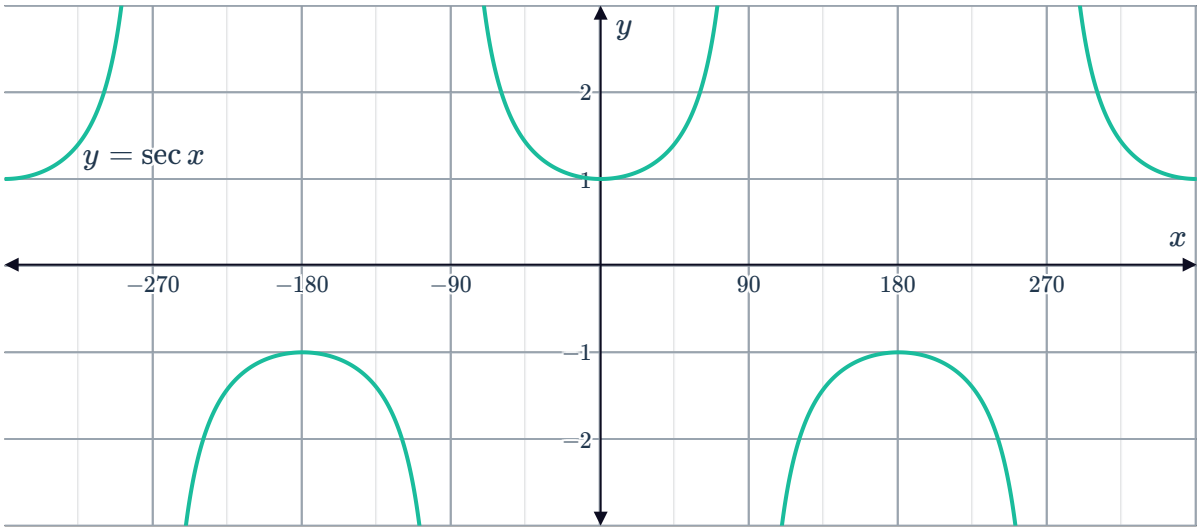
c -1

d



4

a



b 315°

c 360°

d 405°

e $(-45)^\circ$

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5

a

x	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π
$\sec x$	1	$\sqrt{2}$	undefined	$-\sqrt{2}$	-1	$-\sqrt{2}$	undefined	$\sqrt{2}$	1

b $(-\infty, -1] \cup [1, \infty)$

Cosecant function

6

a

x	$-\frac{2\pi}{3}$	$-\frac{\pi}{2}$	$-\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$
$\csc x$	-1.155	-1	-1.414	1.155	1	1.414

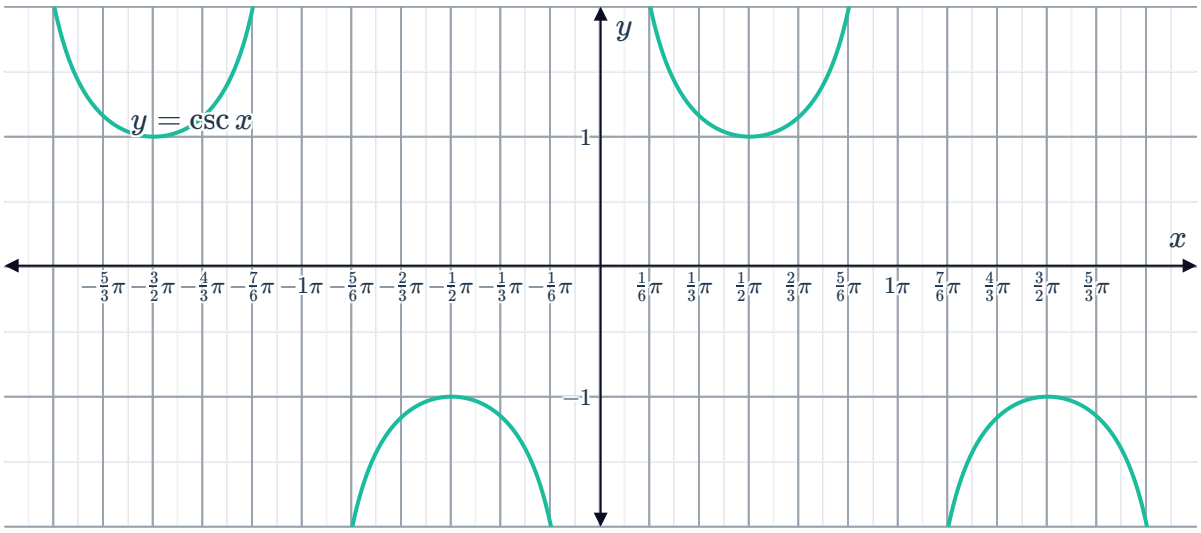
b $x = -2\pi,\ -\pi,\ 0,\ \pi,\ 2\pi$

c $x = -\frac{3\pi}{2},\ \frac{\pi}{2}$

d $x = -\frac{\pi}{2},\ \frac{3\pi}{2}$

e 2π

f



g It does not have a y-intercept since $\csc x$ is not defined at $x = 0$.

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7

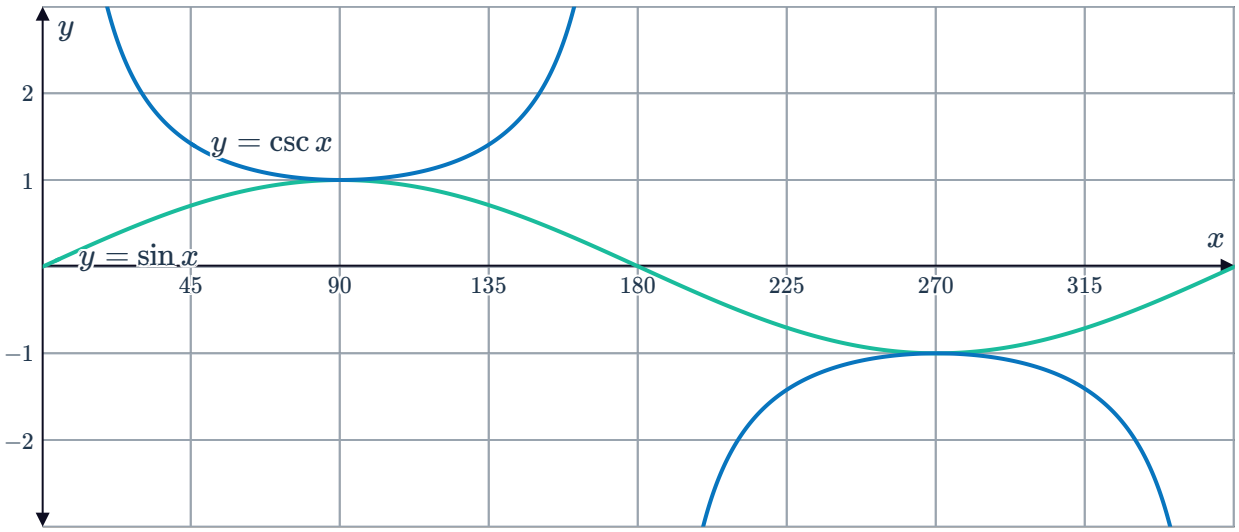
a

x	0°	45°	90°	135°	180°	225°	270°	315°	360
$\sin x$	0	$\frac{1}{\sqrt{2}}$	1	$\frac{1}{\sqrt{2}}$	0	$-\frac{1}{\sqrt{2}}$	-1	$-\frac{1}{\sqrt{2}}$	0
$\csc x$	$-$	$\sqrt{2}$	1	$\sqrt{2}$	$-$	$-\sqrt{2}$	-1	$-\sqrt{2}$	$-$

b 1

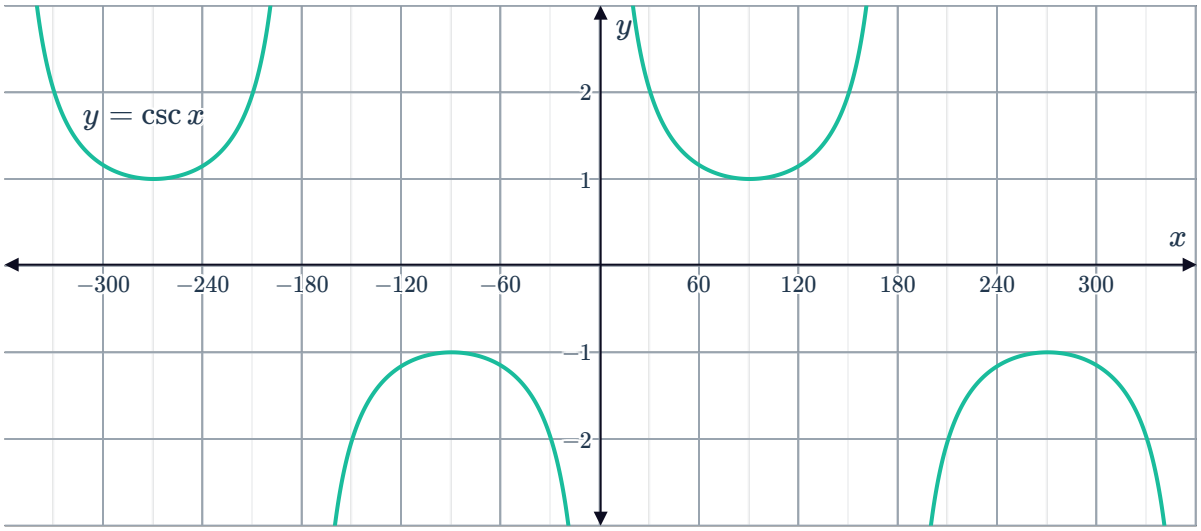
c -1

d



8

a



b 150°

c 360°

d 390°

e -210°

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9

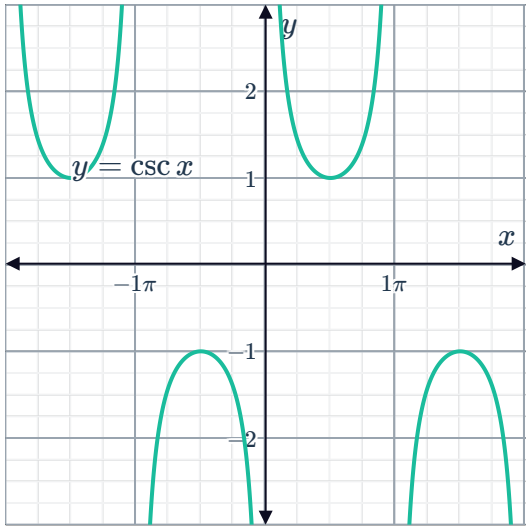
a $x = 0,\ \pi,\ 2\pi$

b

x	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π
$\csc x$	undefined	$\sqrt{2}$	1	$\sqrt{2}$	undefined	$-\sqrt{2}$	-1	$-\sqrt{2}$	undefined

c $(-\infty, -1] \cup [1, \infty)$

d



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Cotangent function

10

a

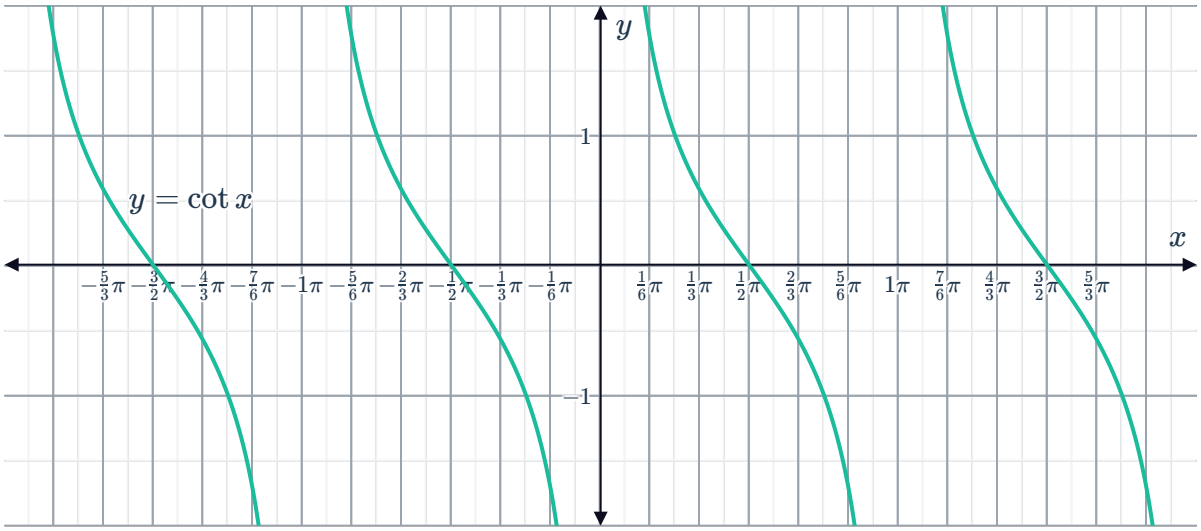
x	$-\frac{3\pi}{4}$	$-\frac{\pi}{2}$	$-\frac{\pi}{6}$	$\frac{\pi}{3}$	$\frac{\pi}{4}$	$\frac{5\pi}{6}$
$\cot x$	1	0	-1.732	0.577	1	-1.732

b $x = -2\pi,\ -\pi,\ 0,\ \pi,\ 2\pi$

c $x = -\frac{3\pi}{2},\ -\frac{\pi}{2},\ \frac{\pi}{2},\ \frac{3\pi}{2}$

d π

e



f Decrease

11

a The asymptotes of one graph correspond to the x-intercepts of the other and vice versa.

b On either graph, the period can be found by finding the distance between successive x-intercepts, or by finding the distance between successive asymptotes.

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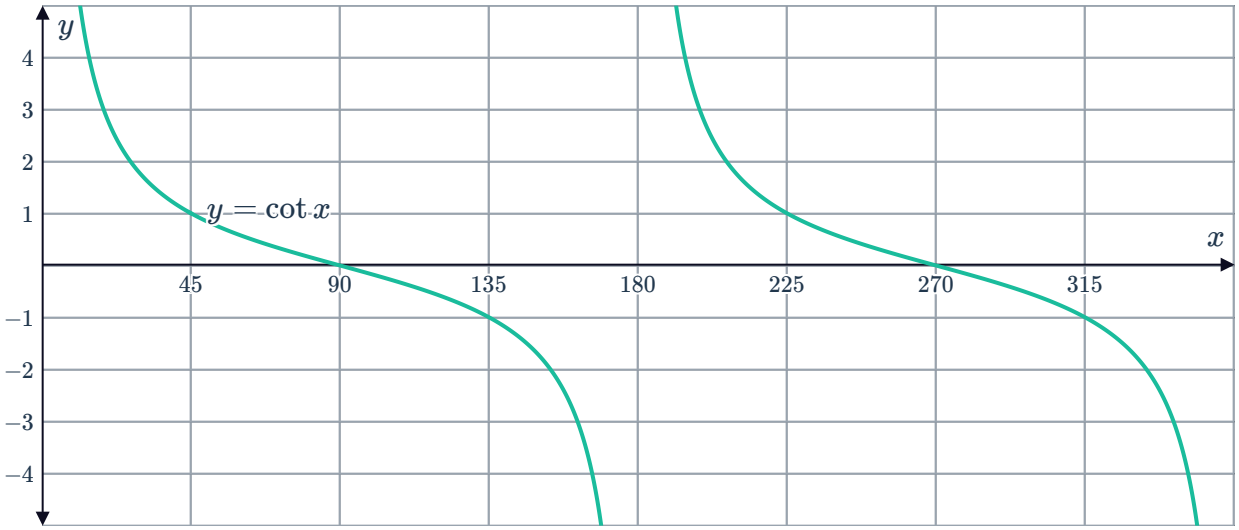
12

a

x	0°	45°	90°	135°	180°	225°	270°	315°	360°
$\cot x$	-	1	0	-1	-	1	0	-1	–

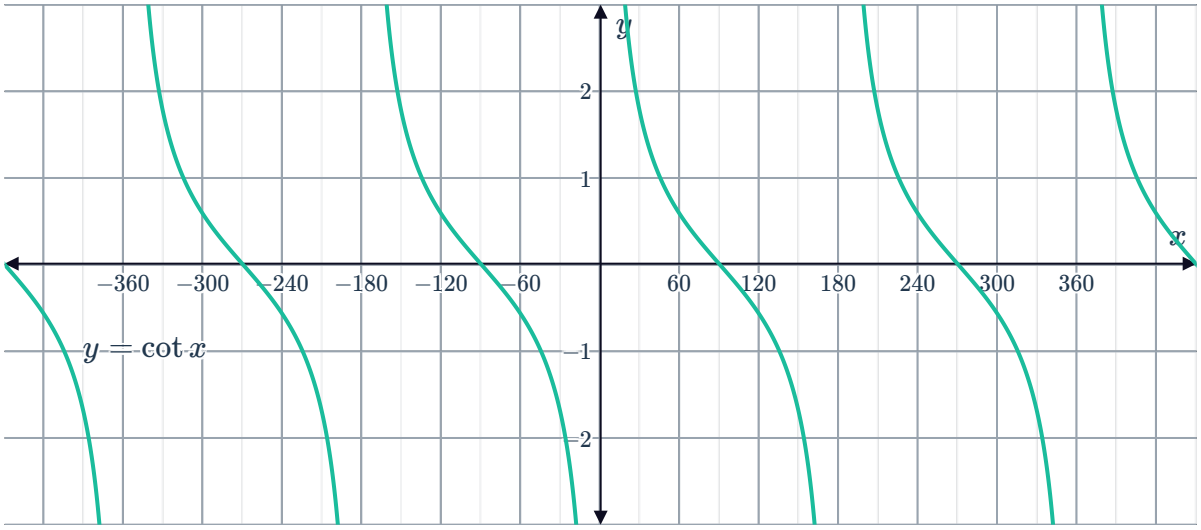
b $90^\circ, 270^\circ$

c



13

a



b 210°

c 180°

d 390°

e -150°

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14

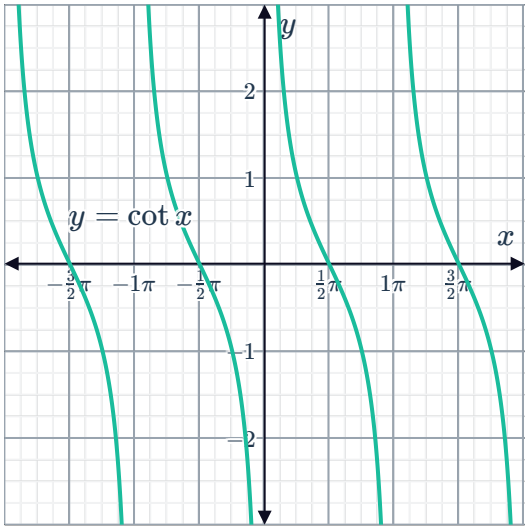
a $x = 0,\ \pi,\ 2\pi$

b

x	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π
$\cot x$	undefined	1	0	-1	undefined	1	0	-1	undefined

c $x = \frac{\pi}{2},\ \frac{3\pi}{2}$

d



e The graph is periodic. It repeats every π radians. Vertical asymptotes occur at $0, \pm\pi, \pm 2\pi \ldots$ and x-intercepts occur at $\pm\frac{\pi}{2}, \pm\frac{3\pi}{2}, \pm\frac{5\pi}{2} \ldots$

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